

Positivity of the determinant maximum

MathlibAnnex.DeterminantFrame.determinantMaximum_pos

theorem

A uniformly scaled dual basis supplies a positive comparison determinant.

Statement

Let $n \in \mathbb{N}$, let E be a finite-dimensional real normed vector space, and fix an ordered basis $b = (b_j)_{j=1}^n$. Write $C_b : E \rightarrow \mathbb{R}^n$ for the coordinate isomorphism, defined by $x = \sum_{j=1}^n (C_b x)_j b_j$, and give its coordinate space the sup norm $\|c\|_\infty = \max_j |c_j|$ (zero in dimension zero). A frame is an ordered family $F = (\ell_i)_{i=1}^n$ of continuous real linear functionals, with no independence assumed. Its matrix and evaluation map are

$$M_F = (\ell_i(b_j))_{i,j}, \quad T_F x = (\ell_i(x))_i = M_F C_b x.$$

The admissible frames form the product of closed dual unit balls

$$\mathcal{U} = \{F : \|\ell_i\|_{E^*} \leq 1 \text{ for every } i\}.$$

The norm on E need not be a coordinate sup norm. Let F_* be an absolute maximizing frame and put $D_b = |\det M_{F_*}|$. Then $D_b > 0$.

Assumptions

Only the finite-dimensional real normed space and the specified basis b are inputs. Dimension zero is allowed. The maximum is taken over the closed dual unit balls, not over unit spheres.

Conclusion

The attained absolute maximum is strictly positive. Consequently a maximizing frame has nonzero determinant, but it need not be unique.

Proof route

Exhibit an admissible scaled dual basis and compare its positive determinant with the attained maximum.

▼ Proof steps

- 1. Scale all coordinate functionals by the same amount.** Let $e_i^* : E \rightarrow \mathbb{R}$ be the dual coordinate functional $e_i^*(x) = (C_b x)_i$ and set

$$S_b = \sum_i \|e_i^*\|, \quad \alpha_b = (1 + S_b)^{-1} > 0, \quad F_0 = (\alpha_b e_i^*)_i.$$

These are the coordinate functionals, sum of their norms, and common scaling factor. Since every term of S_b is nonnegative and $\|e_i^*\| \leq S_b$,

$$\|\alpha_b e_i^*\| = \frac{\|e_i^*\|}{1 + S_b} \leq \frac{S_b}{1 + S_b} \leq 1.$$

Thus admissibility of the scaled coordinate frame applies to this same F_0 .

2. Insert the comparison value into the maximum. The basis identity $e_i^*(b_j) = \delta_{ij}$ gives

$$M_{F_0} = \alpha_b I, \quad |\det M_{F_0}| = \alpha_b^n > 0.$$

Use the scaled coordinate matrix and its determinant, then the maximizing comparison with $F_0 \in \mathcal{U}$:

$$0 < \alpha_b^n = |\det M_{F_0}| \leq |\det M_{F_*}| = D_b.$$

When $n = 0$, $S_b = 0$, $\alpha_b = 1$ and the empty determinant is 1, so the same inequality proves the result.

The proof uses the positive scale, the nonnegative row-size sum, each row norm bounded by that sum, the scaled row and the scaled coordinate frame. These are proof-context definitions, not extra arguments in the exact signature. Their common scalar is α_b , and each row evaluates as $\alpha_b e_i^*(x)$.

Main citations

- coordinate functionals — `MathlibAnnex.DeterminantFrame.rowCoordinateRow`
- sum of their norms — `MathlibAnnex.DeterminantFrame.coordinateRowSize`
- common scaling factor — `MathlibAnnex.DeterminantFrame.coordinateScale`
- admissibility of the scaled coordinate frame —
`MathlibAnnex.DeterminantFrame.coordinateFrame_mem`
- the scaled coordinate matrix — `MathlibAnnex.DeterminantFrame.frameMatrix_coordinateFrame`
- its determinant — `MathlibAnnex.DeterminantFrame.frameDeterminant_coordinateFrame`
- the maximizing comparison —
`MathlibAnnex.DeterminantFrame.abs_frameDeterminant_le_maximizingFrame`
- the positive scale — `MathlibAnnex.DeterminantFrame.coordinateScale_pos`
- the nonnegative row-size sum — `MathlibAnnex.DeterminantFrame.coordinateRowSize_nonneg`
- each row norm bounded by that sum —
`MathlibAnnex.DeterminantFrame.norm_rowCoordinateRow_le_size`
- the scaled row — `MathlibAnnex.DeterminantFrame.coordinateRow`
- the scaled coordinate frame — `MathlibAnnex.DeterminantFrame.coordinateFrame`
- Exact declaration and proof — `MathlibAnnex.DeterminantFrame.determinantMaximum_pos`

Lean source signature (exact)

```
/-- The basis-dependent determinant maximum is positive. In dimension zero,
`coordinateRowSize = 0`, the empty determinant is `1`, and the same proof applies. -/
theorem determinantMaximum_pos (b : Basis (Fin n) ℝ E) :
  0 < determinantMaximum b
```

The explicit input is the basis; the theorem proves positivity of the maximum defined from that basis.

In the source	Mathematical meaning
<code>n; b : Basis (Fin n) ℝ E</code>	A nonnegative integer n and the ordered basis $(b_j)_{j=1}^n$ of E .
<code>[NormedAddCommGroup E] [NormedSpace ℝ E] [FiniteDimensional ℝ E]</code>	The ambient E is a finite-dimensional real normed vector space, with its given norm (not necessarily a coordinate sup norm). These hypotheses occur in the separate surrounding excerpt.
<code>Fin n → ℝ</code>	The coordinate space $\mathbb{R}^n$ with the sup norm. Lean indexes the coordinates by $0, \dots, n-1$; the formulas here use $1, \dots, n$.
<code>Frame n E</code>	The ordered families $F = (\ell_i)_{i=1}^n$ with $\ell_i : E \rightarrow \mathbb{R}$ continuous and real linear. No independence is part of this type.
<code>unitFrameSet n E</code>	The set $\mathcal{U} = \{F : \ \ell_i\ _{E^*} \leq 1 \text{ for every } i\}$. Its elements are families of functionals, not points of E .
<code>maximizingFrame b</code>	The selected absolute maximizing frame F_* in the admissible set.
<code>determinantMaximum b</code>	The real number $D_b = \max_{F \in \mathcal{U}} \det(\ell_i(b_j))_{i,j} $, using the specified basis b .
<code>0 < determinantMaximum b</code>	The conclusion is $0 < D_b$. There is no extra hypothesis that a varying frame is independent or has positive determinant.

The bracketed hypotheses describe E ; they are not conditions on a new frame. The theorem includes $n = 0$.

Names from the source comment and proof context:

In the source	Mathematical meaning
<code>coordinateRowSize b; coordinateScale b</code>	In the source comment and proof context, $S_b = \sum_i \ e_i^*\ _{E^*}$ and $\alpha_b = (1 + S_b)^{-1}$, where $e_i^*(x) = (C_b x)_i$.
<code>coordinateFrame b</code>	The comparison frame $F_0 = (\alpha_b e_i^*)_i$; its matrix is $\alpha_b I$. These named objects are used in the proof, not additional theorem inputs.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 47–49:

```
variable {n : ℕ}
variable {E : Type u} [NormedAddCommGroup E] [NormedSpace ℝ E]
[FiniteDimensional ℝ E]
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 37–49. The full original context is retained with the source evidence.

▼ **Exact content identity**

Declaration: `MathlibAnnex.DeterminantFrame.determinantMaximum_pos`

Accepted content SHA-256: `88c7250e9ce9f5c0ca4075df9465864b85e3b171bda1c39334633047151abe4f`

Accepted source guide SHA-256: `22c18235cae3d2c2b36ea48aa250e62c3b1302f0b08e2b3a55449b759e4dae95`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`